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Building sustainable federated learning models in Fog-enabled UAV-as-a-Service for aerial image classification

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ABSTRACT

The Fog-enabled UAV-as-a-Service (FU-Serve) platform leverages distributed fog nodes to enable real-time data processing for multiple concurrent applications. However, the computational limitations of these fog nodes significantly hamper the execution of resource-intensive deep learning (DL) algorithms, compromising both operational performance and energy sustainability. To address these challenges, we integrated Federated Learning (FL) within the FU-Serve platform, coupled with the development of three specialized FL-based models tailored for on-device image classification. First, we introduced a sustainable adaptation of MobileNetV2 that synergizes Transfer Learning (TL) with FL principles. This model achieves 97.68% accuracy with an 8.64 MB footprint by distributing pre-trained weights to optimize bandwidth efficiency. To further address resource constraints of fog nodes, we designed FUSERNet—a lightweight DL architecture employing separable convolutions and skip connections, which reduces computational overhead while preserving critical feature representations. This model achieves 97.47% accuracy with an ultra-compact footprint of 237 KB, demonstrating a 98.59% reduction in size compared to state-of-the-art models. Finally, our third model, FusionNet, combines the strengths of MobileNetV2 and FUSERNet to deliver a balanced solution, achieving 97.75% accuracy with moderate resource requirements (8.86 MB). We evaluated our models on the AIDER and NDD disaster response datasets, our models demonstrate superior performance in classifying critical natural disaster scenarios. Notably, FusionNet matches SOTA accuracy levels while reducing memory consumption by 50%, and FUSERNet's 0.23 MB size enables deployment on even the most resource-constrained UAVs. These contributions enhance the FU-Serve platform's real-time decision-making capabilities, balancing computational efficiency and mission-critical accuracy for sustainable disaster response.

1. Introduction

Unmanned Aerial Vehicles (UAVs) are widely used in various sectors, including civilian, industrial, military, and disaster relief operations. In disaster relief, UAVs are crucial for rapid aerial surveys, search and rescue missions, and delivering supplies to inaccessible areas [1]. On the other hand, the technological advancements in camera sensors enable these UAVs to detect and analyze disasters such as collapsed buildings, fires, floods, and traffic accidents effectively [2]. Traditionally, the usage of UAVs is single-user-centric in serving different applications, such as disaster relief operations. Such operations require direct deployment, operating, and managing the procured UAVs, which are difficult activities for an end-user. To address these issues, Yapp et al. [3] introduced the UAV-as-a-Service (UaaS) model to provide UAV services to the end user. Further, Pathak et al. [4] proposed the theoretical model of UaaS while describing the detailed architecture and the roles of the actors associated with UaaS. However, traditional UaaS platforms, which rely heavily on cloud computing for data processing, face latency challenges [5] that are detrimental to time-critical applications such as disaster relief operations. Imandi et al. [6] developed the FU-Serve platform, integrating fog computing to reduce latency. The FU-Serve platform enables UAVs to process data onboard, thus decreasing reliance on cloud systems [7].

To enhance emergency response and disaster management, FU-Serve utilizes deep learning (DL) algorithms, specifically Convolutional Neural Networks (CNNs), for real-time disaster image classification. However, deploying DL algorithms within FU-Serve faces challenges due to UAVs' limited computational resources and power constraints [8]. Traditional DL models also require extensive data transfers to centralized servers, increasing bandwidth usage and raising privacy concerns [9].

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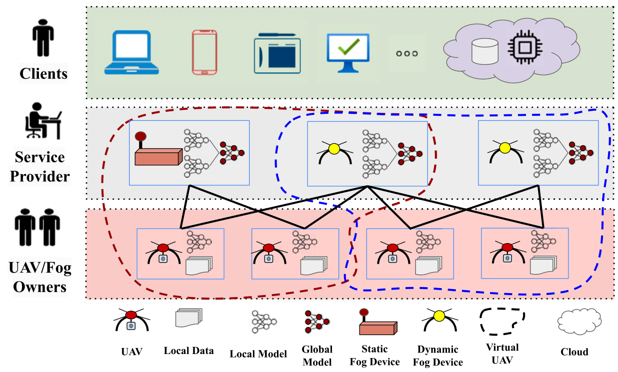


Fig. 1. Integration of FL in the FU-Serve platform.

To overcome these issues, FL has been integrated into the FU-Serve platform to support distributed DL model training across UAVs [10]. FL allows each UAV to process data locally and share only model updates, preserving data privacy and reducing bandwidth needs—ideal for UAV operations that demand real-time analysis and high data security [11]. Additionally, FL promotes onboard data processing, which minimizes latency and decreases reliance on centralized servers, which are key requirements for effective emergency response. FL integration enables scalable, efficient, and secure DL deployment across UAV networks, addressing critical needs for energy efficiency and operational sustainability in resource-constrained environments [12].

Concurrently, TL presents a powerful method for image classification. TL utilizes pre-trained models to apply learned representations to new, related tasks, thus conserving data and reducing computational demands [13]. TL not only improves performance but also expedites the adaptation process, proving invaluable across various image classification contexts. We explore the integration of TL into the FL framework to enhance aerial image classification, consistently surpassing existing state-of-the-art models in disaster scene analysis. By combining FL with TL, we offer a robust solution tailored for scenarios that prioritize privacy, immediate responsiveness, and model accuracy. This proposed approach optimizes UAV-based disaster response systems by enabling UAVs to collaboratively learn and adapt in real-time without compromising data privacy. Utilizing pre-trained models accelerates the learning process and improves responsiveness. Such synergy significantly enhances the UAVs' ability to swiftly recognize and respond to varied disaster scenarios, thereby boosting operational efficiency and effectiveness under critical conditions.

This paper focuses on onboard aerial image classification to automatically assign semantic labels to aerial images captured by UAVs. As illustrated in Fig. 1, the FU-Serve platform utilizes FL across UAVs, where some UAVs act as dynamic fog devices to aggregate model weights, while others serve as FL clients focused on local data processing. This strategic deployment allows dynamic devices to manage model updates, refining the global model by averaging weights from client UAVs. These updates are redistributed for continuous model enhancement. On the other hand, Static fog devices supplement this network by providing robust ground-based aggregation points that enhance the overall processing power and data integrity. The system's design ensures operational flexibility and resilience, enabling asynchronous updates and maintaining function under variable connectivity, which is crucial in disaster-stricken areas. By leveraging both dynamic and static fog nodes, the FU-Serve platform efficiently meets the computational demands of UAV-supported disaster management, ensuring data security and minimizing bandwidth use.

We envision UAVs following predetermined paths, analyzing frames from onboard cameras in real-time. We aim to develop a suite of DL models that optimally balance accuracy, performance, and sustainability, suitable for operation on embedded UAV hardware or mobile control stations. The first model, Modified MobileNetV2, utilizes a transfer learning approach to reduce model size while maintaining high performance, adapting pre-trained model weights for efficient use across fog nodes. This approach not only ensures effective processing but also promotes energy efficiency and reduces bandwidth consumption, aligning with sustainable computing practices. The second model, FUSERNet, is a compact yet effective model for aerial image classification, specifically designed to facilitate efficient distributed learning across UAV networks. It achieves exceptional accuracy in time-critical applications while incorporating energy-efficient design elements such as separable convolutions and skip connections. These features enhance performance and preserve critical information, making FUSERNet particularly suitable for sustainable operation in complex disaster situations. Building upon these, our third model, FusionNet, integrates the robust features of both Modified MobileNetV2 and FUSERNet to provide a balanced solution with enhanced accuracy and moderate model size. These proposed models facilitate automated monitoring and inspection, enhancing preparedness and situational awareness during emergency responses. Each model uniquely contributes to our goal of improving decision-making capabilities during time-critical operations using the FU-Serve platform.

1.1. Motivation

The FU-Serve platform, which leverages fog nodes to process and access multiple real-time applications, uses deep learning algorithms to enhance emergency response and disaster management. However, the platform faces different challenges, such as the large size of deep learning models exceeding the computational limits of fog devices, privacy risks, and extensive bandwidth requirements associated with data transmission from UAVs to centralized servers. These challenges are critical in disaster scenarios where a compromised communication infrastructure significantly impacts service delivery within the stipulated time. Additionally, these issues emphasize the need for sustainable solutions that minimize energy consumption and optimize resource usage. On the other hand, the FU-Serve platform features a service-oriented architecture that caters to varying client expectations, such as uninterrupted and timely services with high accuracy. Typically, FU-Serve architecture includes three different types of UAVs—F-UAVs, H-UAVs, and N-UAVs. An F-UAV is equipped with high-end fog services but lacks sensing capabilities, making them suitable for processing moderate-sized models like Modified MobileNetV2 and FusionNet, which are optimized for sustainability by balancing accuracy and computational efficiency. H-UAVs offer both sensing and fog computing capabilities but have limited memory and processing power, which makes them ideal for running lightweight models such as FUSERNet. N-UAVs, on the other hand, lack fog computing capabilities altogether.

The existing literature fails to address these specific challenges fully, especially the model size, privacy concerns, and sustainability issues inherent in the FU-Serve platform. This gap highlights the urgent need for developing tailored deep-learning models that align with the distinct computational and operational capacities of the different UAV types within the platform. For instance, higher accuracy requirements from certain clients can be met by deploying FusionNet on F-UAVs, while cost-sensitive operations might prefer the more economical H-UAVs equipped with FUSERNet. Developing these sustainable models is crucial for ensuring efficient and secure time-critical applications such as disaster management operations.

1.2. Contributions

Our research enhances onboard aerial image classification for disaster response within the Fog-enabled UAV-as-a-Service (FU-Serve) platform. Below are the key contributions of our work:

- **MobileNetV2 Optimization:** We began by optimizing MobileNetV2, chosen for its computational efficiency—a crucial attribute for UAV applications. Despite its benefits, the model's substantial size was unsuitable for compact UAVs with limited memory. To address this, we modified MobileNetV2 by freezing early layers to utilize pre-trained features and fine-tuning later layers specifically for aerial disaster image classification. This modification resulted in a robust model that not only achieves an impressive accuracy of 97.68% but also maintains a compact size of 8.64 MB.
- **Development of FUSERNet:** Building on this success, we developed FUSERNet to further reduce the model size to better fit the limited computational resources of UAVs. FUSERNet, a compact CNN, leverages separable convolutions to efficiently filter inputs and reduce overhead and integrates skip connections to enhance training depth and data retention. This architecture significantly reduces the model size to just 237 KB – 34% smaller than MobileNetV2 – making it exceptionally suitable for real-time applications on low-power UAV devices, achieving an accuracy of 97.47%.
- **Creation of FusionNet:** Capitalizing on the strengths of Modified MobileNetV2 and FUSERNet, we introduced FusionNet, a hybrid model that merges the robust features of its predecessors. FusionNet maintains a modest model size of 8.86 MB and improves accuracy to 97.75%, demonstrating the advantages of architectural synergy in boosting performance while ensuring manageability for UAV deployment in time-critical scenarios.
- **Model Validation and Testing:** Finally, the efficacy of our proposed models was rigorously tested through simulation experiments involving UAVs equipped with fog computing capabilities. We utilized the AIDER and NDD datasets for validation, creating a comprehensive testing environment that assessed model performance across various disaster scenarios. These experiments confirmed the models' real-time processing abilities and reliability, affirming their viability and effectiveness for practical deployment within the FU-Serve platform.

2. Related works

This section reviews the literature relevant to our proposed work while focusing on two aspects—(i) UaaS Platform and (ii) FL and TL-based image classification.

2.1. UAV-as-a-Service platform

UAVs are increasingly utilized across various domains due to their scalability and adaptability. Traditional UAV operations, however, often involve single-user-centric deployment, leading to challenges such as suboptimal resource utilization during periods of inactivity, limited deployment mechanisms, and the burden on users to manage upgrades independently. To overcome these limitations, Yapp et al. [3] introduced the UaaS paradigm, enabling on-demand provisioning for various applications. Despite its flexibility, UaaS lacks UAV virtualization, limiting its ability to support multiple applications concurrently. Therefore, Pathak et al. [4] provided a mathematical foundation supporting the importance of UAV virtualization. In the proposed work, the authors discussed different components and the actors of a UaaS platform. In a UaaS platform, different heterogeneous UAVs collaboratively serve an application.

However, the selection of a UAV is a crucial task. Considering the same, Roy et al. [14] introduced the Opti-U scheme to select an optimal UAV while focusing on serving long-endurance missions. The traditional UaaS platform is based on traditional cloud architecture, where the service latency is significant and unacceptable for time-critical applications. To address the time-criticality issues, Imandi et al. [6] introduced the FU-Serve platform. The FU-Serve platform employs static and dynamic fog devices to handle time-critical applications. Additionally, the FU-Serve platform supports heterogeneous UAVs and fog devices. This enables the adoption of FL into the FU-Serve platform to enhance platform intelligence.

2.2. FL and TL-based image classification

Traditional DL algorithms rely on centralized servers, requiring data transfer for real-time analysis, resulting in significant bandwidth consumption and privacy concerns. The deployment of FL in UAV networks introduces a paradigm shift away from traditional centralized server-based systems, primarily targeting the crucial challenges of bandwidth consumption and data privacy. Despite FL's advantages in decentralization, its application in UAV networks, especially for real-time tasks like on-device image classification, has emphasized certain operational challenges. These challenges include inefficiencies concerning model size, computational demands, and the need for adaptability in dynamic environments, which are critical in disaster management scenarios.

Federated Learning's ability to enhance data privacy and reduce bandwidth usage has been well-noted across various domains. Bekbulatova et al. [15] highlight the privacy and bandwidth challenges inherent in traditional DL systems, which rely on centralized data processing—a method impractical for UAVs in field operations where both data security and minimal bandwidth are essential. The benefits of FL's decentralization have also been demonstrated in fields like agriculture by Friha et al. [16], and healthcare by Antunes et al. [17]. However, these studies do not address the high mobility and variable network conditions typical in UAV networks, pointing to a gap in context-specific adaptations.

Moreover, the integration of TL in UAV-based applications has seen limited exploration. Kyrkou et al. [18] developed EmergencyNet for efficient disaster image classification on UAVs, showing significant progress. Nonetheless, the performance of their model in the constrained computational settings typical of smaller UAVs has not been fully explored. Reis et al. [19] further demonstrate the potential of TL in quickly adapting to new tasks, such as forest fire detection using pre-trained models like InceptionV3 and DenseNet121. Their work significantly enhances performance but lacks focus on the federated aspect, which further optimizes bandwidth and privacy. Furthermore, Bhadra et al. [20] emphasize the significance of customized TL techniques in enhancing UAV image classification, where their MFEMANet model leverages a multi-branch feature extraction mechanism with a mixed-attention approach to improve the semantic interpretation of disaster-affected images significantly. This approach highlights the potential of TL to refine feature extraction processes essential for high-fidelity classification in complex, dynamic environments typical of UAV applications. Although the paper by Bashir et al. [21] discusses an efficient CNN-based classification system for disaster events using UAV-aided images, the considerable size of the model poses a deployment challenge on UAVs, which typically require more compact, computationally affordable solutions due to their limited onboard processing capabilities.

Additionally, the need for advanced DL techniques for efficient feature representation in UAV-based operations is emphasized by Hadid et al. [22], who integrated deep learning with singular value decomposition. While their approach highlights the importance of efficient feature representation, crucial for resource-constrained UAVs, it does not incorporate the federated learning framework, which enhances model training in distributed UAV networks.

Synthesis: The literature review shows the expansive use of UAVs, particularly in disaster management and rescue operations. The literature also highlights the UaaS platform's effectiveness in IoT applications and its evolution into the FU-Serve platform. The FU-Serve significantly enhanced service delivery in time-critical scenarios through the integration of fog computing. Additionally, the FU-Serve platform allows the classification of ongoing scenes through DL algorithms.

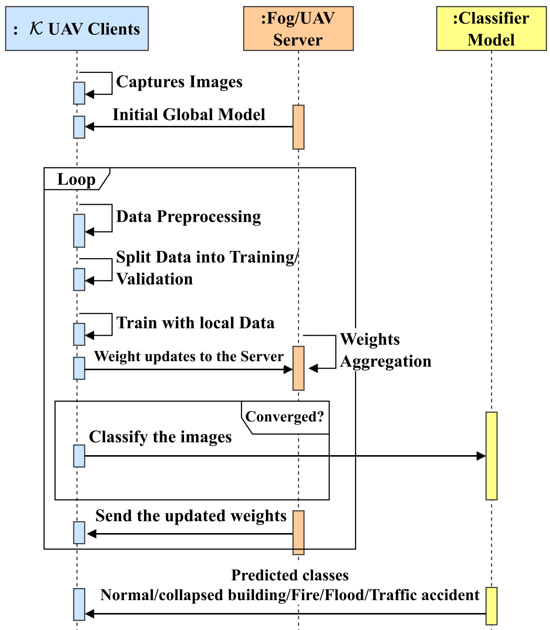


Fig. 2. The proposed DL models workflow.

Despite the advances, the integration of high-performance DL algorithms directly within the FU-Serve platform poses challenges due to significant processing and memory demands. This paper explores the potential of employing compact CNN models and delves into combining federated learning with transfer learning techniques. This integration allows UAVs to quickly adapt to diverse environments using shared insights and pre-trained models, improving accuracy and response times while maintaining data privacy. This streamlined method sets our UAV applications apart in emergency management.

3. Proposed aerial image classification workflow

CNN architectures are popular for classification tasks, notably in high-profile competitions such as the ImageNet Large Scale Visual Recognition Competition (ILSVRC) [23]. In our work, we leverage popular CNN architectures to undertake the task of aerial disaster image classification. The entire workflow is detailed in Fig. 2, starting with UAVs capturing images, followed by data preprocessing to optimize them for the learning process. Given the computationally demanding nature of deep neural networks and their dependence on extensive datasets, we employ an 80%–20% split for training and testing, respectively. This split enhances training efficiency and minimizes bias in the model.

The FL cycle begins with the fog device initiating the global model with pre-trained weights, which are subsequently distributed to a selected group of $\mathcal{K}$ clients as shown in Fig. 2. These clients perform training on their local datasets and then transmit their model updates back to the server. The server employs the Federated Averaging algorithm [24] to merge these updates and continuously assess convergence. Training iteratively continues until convergence is achieved.

For building the global model on the server, we incorporate TL, significantly expediting the training process and curtailing both the duration and loss associated with model training [25]. As depicted in Fig. 3, TL utilizes pre-trained models that have been extensively trained on large-scale datasets, thus eliminating the necessity to train from scratch. This approach facilitates the reuse of complex feature representations learned previously, efficiently adapting these for the nuanced task of classifying aerial disaster images. The detailed steps are shown in Algorithm 1.

Algorithm 1: Federated Learning-based Framework for Image Classification

Input: $\mathcal{D} = \{(x_i, y_i)\}_{i=1}^{\mathcal{N}_k}$ with $x_i \in \mathbb{R}^d$, $y_i \in \{c_1, c_2, c_3, c_4, c_5\}$, Set of UAVs $\mathcal{U} = \{\mathcal{U}_1, \ldots, \mathcal{U}_k\}$, maximum rounds $\mathcal{T}$, learning rate η

Output: Predicted disaster scenario $\hat{y}$

Initialization:

Distribute $\mathcal{D}$ into subsets $\{\mathcal{D}_k\}_{k=1}^{\mathcal{N}_k}$ across UAVs

Initialize local models $\{\theta_k^0\}_{k=1}^{\mathcal{N}_k}$ randomly or with pre-trained weights

Federated Learning Process: for $t = 1$ **to** $\mathcal{T}$ **do**

  Local Training: foreach $k \in \mathcal{U}$ ***in parallel*** **do**

    Perform local training on $\mathcal{D}_k$:

    $\theta_k^{t+1} = \theta_k^t - \eta \nabla_\theta \mathcal{L}_k(\theta_k^t)$

    where $\mathcal{L}_k(\theta) = -\frac{1}{|\mathcal{D}_k|} \sum_{(x,y)\in\mathcal{D}_k} \log f_y(x;\theta)$

  Global Aggregation at Server:

  The central server aggregates model parameters from all UAVs:

  $\theta^{t+1} = \sum_{k=1}^{\mathcal{N}_k} \frac{|\mathcal{D}_k|}{\sum_{j=1}^{\mathcal{N}_k} |\mathcal{D}_j|} \theta_k^{t+1}$ Broadcast updated global model θ^{t+1} to all UAVs.

Prediction: For new input x_{new}, UAVs use the global model for prediction:

$\hat{y} = \operatorname{argmax}_{c\in\{1,\ldots,5\}} f_c(x_{\text{new}}; \theta^{\mathcal{T}})$

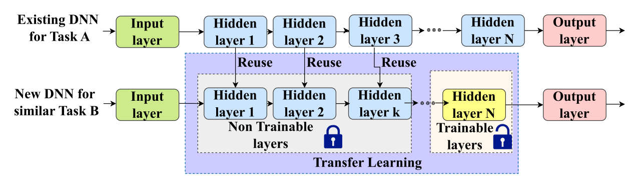


Fig. 3. Implementation of transfer learning on Task B using a pre-trained model from Task A.

We have integrated several state-of-the-art architectures accessed through the Keras API, including VGG16 [26], DenseNet121 [27], InceptionV3 [28], EfficientNetB0 [29], EfficientNetV2B0 [30], NASNet-Mobile [31], SqueezeNet [32], MobileNetV2 [33], and MobileNetV3 [34]. These models are pre-trained on the expansive ImageNet dataset which makes these models suitable for aerial image classification. Training such complex models from scratch is prone to overfitting, particularly given our dataset's limited size relative to these models' complexity. To tailor these models to our specific requirements, we designate certain layers as non-trainable while allowing others to remain trainable. Each network retains its feature extraction components in a frozen state, and all necessary preprocessing steps are applied to the input images. Following established precedents, a classification layer is added on top of these models, typically incorporating a global average-pooling layer before the final dense layers, resulting in a softmax output. This approach reduces the parameter count, thereby minimizing computational and memory demands and ensuring performance comparable to traditional methods employing fully connected layers. Thus, the pre-trained models we utilize exhibit enhanced efficiency in memory usage and computational operations compared to other networks documented in the literature for similar tasks.

3.1. Proposed model enhancements

Our proposed models – MobileNetV2, FUSERNet, and FusionNet – showcase distinct enhancements that redefine UAV capabilities, setting them apart from conventional architectures like VGG16, DenseNet121, and InceptionV3. We began by optimizing MobileNetV2 for computational efficiency to suit UAV constraints, modifying it through the

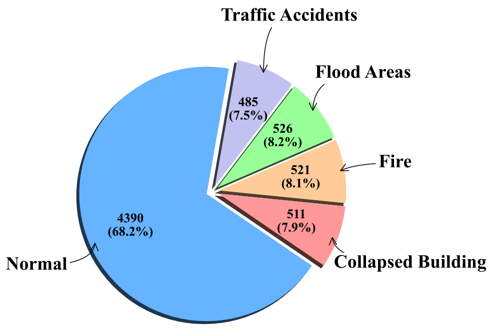


Fig. 4. Summary of AIDER dataset.

Fig. 5. Sample images from NDD dataset.

freezing of early layers and fine-tuning of later layers specifically for aerial disaster image classification. This adaptation significantly reduced its size, making it highly suitable for UAVs with limited memory capacities. Building on this foundation, we developed FUSERNet, a compact CNN that utilizes separable convolutions to minimize computational load. FUSERNet also integrates skip connections to enhance training efficacy and data retention, achieving a substantial reduction in model size. FusionNet merges the strengths of both MobileNetV2 and FUSERNet, leveraging architectural synergies to enhance performance without compromising efficiency, making it ideal for deployment in UAVs operating under critical constraints.

3.2. Dataset details and preprocessing:

The AIDER dataset [18] is pivotal for our aerial image classification due to its comprehensive and multiclass nature, making it a unique tool for disaster response training. Unlike specific disaster image databases such as FloodNet [35], which focuses on flooding with 2343 images for tasks like semantic segmentation and classification, or Instance Segmentation in Building Damage Assessment (ISDBA) [36], which comprises 1030 images mainly for instance segmentation of hurricane and flood impacts, the AIDER dataset offers a broader scope. It includes five diverse classes: collapsed buildings, fire, flooded areas, traffic accidents, and a normal class. Sourced from various online platforms including Google Images, Bing Images, YouTube, and news agencies, the AIDER dataset provides a robust challenge due to its class imbalance, with the normal class significantly outnumbering the disaster-specific classes. This dataset comprises around 500 images per disaster class and 4390 images for the normal class, which is depicted in Fig. 4. To counter the imbalance and enhance the model's ability to generalize, we employed data augmentation techniques such as random flipping and rotation during dataset preparation. The AIDER dataset's accessibility via the Zenodo portal ensures it is available for noncommercial and academic use, facilitating widespread research application.

In addition to AIDER, we also utilized the NDD (Natural Disaster Dataset), which provides a balanced classification challenge with four disaster categories: cyclones (700 images), earthquakes (1013 images), floods (805 images), and wildfires (808 images). Each class in the NDD is carefully curated to represent distinct disaster scenarios, providing a well-rounded dataset for testing the efficacy of UAV-based disaster response systems. Fig. 5 illustrates few samples from each category of the NDD, providing a visual representation of the dataset's diversity. Similar to our processing of the AIDER dataset, the NDD underwent extensive preprocessing to standardize image sizes and implement normalization, ensuring consistent input quality for our models. The NDD's comprehensive coverage of various natural disasters makes it an invaluable resource for developing and refining UAV capabilities in real-time emergency management. These datasets are instrumental in developing our UAV-based aerial image classification models, allowing for extensive testing and optimization across a wide range of disaster types and conditions.

4. Proposed models

4.1. Federated transfer learning for FU-Serve

Among the various available CNN architectures evaluated for aerial disaster image classification, MobileNetV2 was chosen as the base model due to its efficiency in computational resource utilization, making it particularly suitable for environments with limited computational capacity. In our customized application of MobileNetV2, the initial 48 layers are frozen to capitalize on the pre-trained features while allowing the later layers to adapt to the specifics of our aerial imagery dataset. Integrating a global average pooling layer is designed to reduce the dimensionality of the feature maps efficiently. This layer performs an averaging operation over each feature map and is defined mathematically as:

$$\text{Global Average Pooling} = \frac{1}{H \times W} \sum_{i=1}^{H} \sum_{j=1}^{W} x_{ij} \tag{1}$$

where H and W are the dimensions of the feature map, and x_{ij} is the value at the (i, j) position of the feature map.

Subsequent to the pooling layer, a dense layer is added, characterized by its parameters: trainable weights (w) and biases (b). The output of this layer, denoted as z, is computed via the linear transformation:

$$z = w \cdot x + b \tag{2}$$

where x is the input vector to the dense layer. This output is then passed through a sigmoid activation function to achieve a probabilistic classification output:

$$h_{wb}(z) = \frac{1}{1 + e^{-z}} \tag{3}$$

Fig. 6 delineates the transfer-learning-enhanced MobileNetV2 architecture, which has demonstrated exemplary performance, achieving a 97.68% accuracy with a model size of 8.64 MB. Despite its effectiveness, the considerable size of Modified MobileNetV2 presents deployment challenges on resource-constrained UAVs. To mitigate this, we developed FUSERNet, a significantly more compact CNN architecture, which occupies only 237 KB—a reduction by a factor of 34 compared to MobileNetV2, albeit with a slight decrement in accuracy by 0.21%, resulting in 97.47%. These results are obtained on the AIDER dataset, highlighting the practical applicability of our model in time-critical scenarios. Detailed discussion on FUSERNet and its architecture is presented in the following section (Section 4.2).

4.2. The proposed novel lightweight model—FUSERNet

The proposed model, FUSERNet, integrates multiple advanced concepts within CNN, tailored explicitly for efficient aerial disaster image

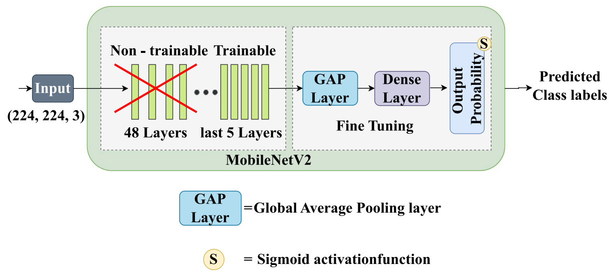


Fig. 6. The proposed modified MobileNetV2 architecture.

classification. As depicted in Fig. 7, the model starts with an Input Layer, $\mathbb{R}^{H\times W\times C}$, where H, W, and C denote the height, width, and number of channels of the input image, respectively. This layer serves as the foundational entry point for input data. Following the input layer, the model incorporates a Convolution Block. The mathematical expression for this block is:

$$x = \text{ReLU}(\text{BN}(\text{Conv}(x, K, F, S))) \tag{4}$$

where x is the input image tensor of the shape $\mathbb{R}^{H\times W\times C}$, $\text{Conv}(x, K, F, S)$ denotes the convolution operation applying filters F of size $K \times K$ with stride S, followed by batch normalization (BN) and activation through the rectified linear unit (ReLU). This block is pivotal as it extracts primary feature maps from raw inputs. After the initial convolution, the architecture employs Separable Convolution Blocks, which utilize depthwise followed by pointwise convolutions. These blocks are designed to reduce computational complexity while ensuring the network's capacity to process image features effectively. The operation within these blocks can be articulated as follows:

$$x = \text{ReLU}(\text{BN}(\text{SepConv}(x, F, K))) \tag{5}$$

with $\text{SepConv}(x, F, K)$ symbolizing the separable convolution operation using filters F and kernel size K. A Self-Attention Module is integrated into the architecture to enhance the model's capability to focus on relevant features. This module adjusts the feature channels based on their importance, calculated as:

$$y = \text{Conv}(\text{ReLU}(\text{Conv}(x, \frac{F}{8}, 1))) \tag{6a}$$

$$\text{Attention} = \sigma(\text{Conv}(\text{ReLU}(\text{Conv}(\text{ReLU}(y))))) \tag{6b}$$

where σ denotes the sigmoid activation function, scaling the convolution output to a [0, 1] range, which facilitates effective feature recalibration by element-wise multiplication with the input feature map:

$$\text{Attended Inputs} = x \odot \text{Attention} \tag{7}$$

Skip Connections are implemented to facilitate training and mitigate vanishing gradients, computed as:

$$\text{Skip Output} = \text{Conv}(x, 2F, 1, 2) \tag{8}$$

allowing alternate shortcut paths for gradient flow during backpropagation. The output from the last self-attention module and the corresponding skip connection undergo element-wise addition followed by activation:

$$x = \text{ReLU}(x_{\text{last}} + \text{Skip Output}) \tag{9}$$

Towards the final stages, the model incorporates a Global Average Pooling layer, which significantly reduces each feature map's dimensions to a single value, preparing the model for classification:

$$x = \frac{1}{H \times W} \sum_{h=1}^{H} \sum_{w=1}^{W} x_{h,w} \tag{10}$$

The last layers of the model comprise a Dense Layer for further feature processing and integration:

$$x = \text{ReLU}(\text{BN}(\text{Dense}(x, 256))) \tag{11}$$

and finally, a Softmax Output Layer for classifying the images into predefined categories, ensuring the output probabilities sum to one:

$$\text{Output} = \text{Softmax}(\text{Dense}(x, \text{num\_classes})) \tag{12}$$

This structured flow from input to output ensures a comprehensive encapsulation of features, maintaining robustness and efficiency, particularly suited for time-critical applications.

4.3. Mean prediction for image classification using FusionNet

Our custom-designed deep neural network, FUSERNet, is combined with the transfer-learning-based MobileNetV2 to create the fusion model, FusionNet, as illustrated in Fig. 8. The distinct capabilities of FUSERNet and Modified MobileNetV2 contribute synergistically to the fusion process. FUSERNet, being tailored for specific feature extraction, complements Modified MobileNetV2's broader understanding of complex patterns. The fusion operation involves averaging the predictions from both FUSERNet and Modified MobileNetV2, denoted mathematically as:

$$\text{Fusion Prediction} = \frac{1}{2}(P_{\text{FUSERNet}} + P_{\text{MobileNetV2}}) \tag{13}$$

where P_{FUSERNet} and $P_{\text{MobileNetV2}}$ represent the respective predictions from FUSERNet and Modified MobileNetV2. The proposed approach aims to create a more robust and accurate model. Remarkably, this fusion yielded an accuracy of 97.75%, with only a modest increase in model size compared to Modified MobileNetV2, reaching 8.86 MB on AIDER dataset. This collaborative approach is thoroughly examined in the experimental section, shedding light on the intricacies of model fusion and its impact on classification performance.

4.4. Complexity analysis

In each training round t, every UAV k individually processes its local dataset $\mathcal{D}_k$, where $|\mathcal{D}_k|$ represents the number of data samples, and $\mathcal{M}$ denotes the computational cost associated with a single forward pass through the model. Consequently, the computational complexity for each UAV is given by $\mathcal{O}(|\mathcal{D}_k| \cdot \mathcal{M})$. Given that computational time is primarily dictated by the UAV with the largest dataset, the per-round computational complexity in the worst case scenario becomes $\mathcal{O}(\max_k(|\mathcal{D}_k| \cdot \mathcal{M}))$.

In the proposed FL-based approach, each UAV transmits its updated model parameters θ to a central server, resulting in a communication complexity of $\mathcal{O}(|\theta|)$ per UAV. Subsequently, the central server aggregates these parameters through a weighted averaging method, incurring a computational complexity of $\mathcal{O}(\mathcal{N}_k \cdot |\theta|)$, where $\mathcal{N}_k$ signifies the total number of participating UAVs. Following aggregation, the updated global model parameters are broadcast back to all UAVs, adding another communication overhead of $\mathcal{O}(\mathcal{N}_k \cdot |\theta|)$.

Therefore, the combined complexity per training round, incorporating both local computation and communication overheads, is represented as

$$\mathcal{O}\left(\max_k(|\mathcal{D}_k| \cdot \mathcal{M}) + 2 \cdot \mathcal{N}_k \cdot |\theta|\right) \tag{14}$$

Extending this analysis across $\mathcal{T}$ training rounds, the overall complexity of the FL-based approach, capturing all computational and communication aspects, is expressed as:

$$\mathcal{O}\left(\mathcal{T} \cdot \left(\max_k(|\mathcal{D}_k| \cdot \mathcal{M}) + 2 \cdot \mathcal{N}_k \cdot |\theta|\right)\right) \tag{15}$$

This complexity analysis effectively outlines the computational demands on UAVs, communication overhead, and the aggregation workload within the proposed FL-based FU-Serve platform.

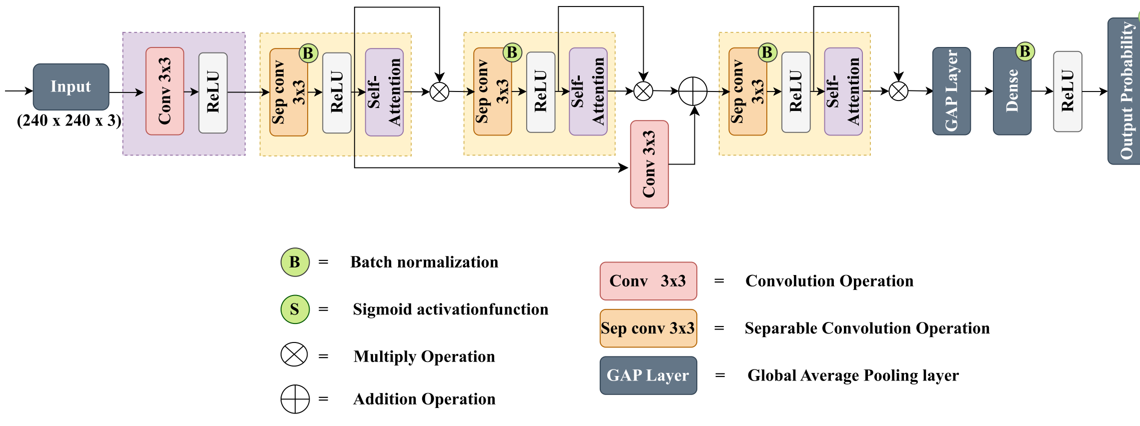


Fig. 7. The proposed light weight FUSERNet architecture.

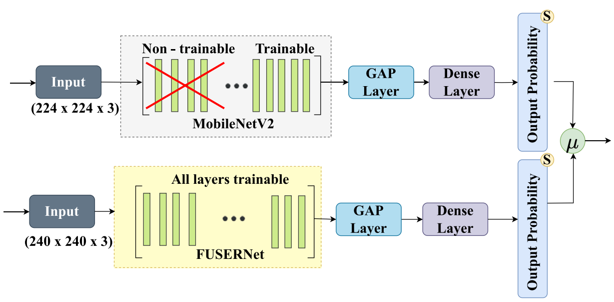


Fig. 8. The proposed FusionNet architecture.

Table 1
Comparative complexity analysis of FUSERNet vs. other models.

Model	Complexity
InceptionV3	$\mathcal{O}(HW(C+F)K^2+F\cdot D)$
VGG16	$\mathcal{O}(HW\cdot 3^2\cdot 3\cdot 64+\cdots+512\cdot D)$
DenseNet121	$\mathcal{O}\left(\frac{HW}{2^2}\sum_{i=1}^{6}(64+32(i-1))3^2\cdot 32+\cdots+1024\cdot D\right)$
EfficientNetB0	$\mathcal{O}(HWC(K^2+F)+F\cdot D)$
NASNetMobile	$\mathcal{O}(HWC(K^2+F)+F\cdot D)$
MobileNetV2	$\mathcal{O}(HWCK^2F+HWCF+F\cdot D)$
Modified MobileNetV2	$\mathcal{O}(HWCK^2F+HWCF+F\cdot D)$
FusionNet (Ours)	$\mathcal{O}(HW[CK^2(F+1)+2CF+F^2]+F\cdot D)$
MobileNetV3	$\mathcal{O}(HWC(K^2+F)+F\cdot D)$
EmergencyNet	$\mathcal{O}(HWCK^2\cdot 32+\cdots+64\cdot D)$
FUSERNet (Ours)	$\boldsymbol{\mathcal{O}(HWCK^2F+HWCF+HWF^2+F\cdot D)}$

4.5. Computational complexity of proposed models vs. Other models

In our complexity analysis, we denote H and W as the height and width of the input feature map, respectively, C as the number of input channels, K as the kernel size of convolutional filters, F as the number of filters, and D as the output dimensionality of fully connected layers. Initially, models such as InceptionV3 involve both standard and factorized convolutions, resulting in a complexity of $\mathcal{O}(HW(C+F)K^2+F\cdot D)$. Similarly, VGG16 employs stacked standard convolutions, yielding $\mathcal{O}(HW\cdot 3^2\cdot 3\cdot 64+\cdots+512\cdot D)$. On the other hand, DenseNet121 incorporates densely connected convolutions across its architecture, which gives a complexity of $\mathcal{O}\left(\frac{HW}{2^2}\sum_{i=1}^{6}(64+32(i-1))3^2\cdot 32+\cdots+1024\cdot D\right)$. Furthermore, models like EfficientNetB0, NASNetMobile, and MobileNetV3 leverage depthwise separable convolutions, resulting in a computational complexity of $\mathcal{O}(HWC(K^2+F)+F\cdot D)$. Additionally, MobileNetV2 and Modified MobileNetV2 explicitly separate their convolutions into depthwise and pointwise operations, expressed as $\mathcal{O}(HWCK^2F+HWCF+F\cdot D)$. FusionNet adds complexity with additional fusion computations, achieving $\mathcal{O}(HW[CK^2(F+1)+2CF+F^2]+F\cdot D)$. In contrast, EmergencyNet simplifies its structure to achieve $\mathcal{O}(HWCK^2\cdot 32+\cdots+64\cdot D)$. Finally, our proposed FUSERNet demonstrates improved computational efficiency, achieving a comparatively lower complexity of $\mathcal{O}(HWCK^2F+HWCF+HWF^2+F\cdot D)$, making it particularly suitable for efficient deployment. Table 1 summarizes these complexities comprehensively.

4.6. Theoretical analysis

Theorem 1. *Given an input feature map of height H and width W, with C input channels, using F filters of size $K\times K$, and assuming a fully connected layer dimensionality D, the implementation of depth-wise separable convolutions in FUSERNet reduces the computational complexity from $\mathcal{O}(HWCK^2F)$ for standard convolutions to $\mathcal{O}(HWCK^2+HWCF)$ for separable convolutions.*

Proof. Standard convolutional operations involve applying each $K\times K$ filter across all C channels to each of the F filters, cumulating in a computational demand of $\mathcal{O}(HWCK^2F)$. In contrast, separable convolutions entail:

- A depth-wise convolution that applies a single $K\times K$ kernel per channel, yielding $\mathcal{O}(HWCK^2)$.
- A subsequent point-wise convolution that combines these depth-wise results across F filters, requiring $\mathcal{O}(HWCF)$.

This structural modification results in a total computational load of $\mathcal{O}(HWCK^2+HWCF)$, thereby significantly economizing on computational resources, particularly beneficial when $F\gg K$. □

Lemma 1. *Skip connections within FUSERNet enhance training stability by facilitating the preservation and direct propagation of gradients through the network layers, effectively mitigating the vanishing gradient problem prevalent in deep architectures.*

Proof. Consider the gradient of the loss function $\mathcal{L}$ with respect to the weights at layer l, W_l, within a traditional deep learning framework:

$$\frac{\partial\mathcal{L}}{\partial W_l}=\prod_{i=l+1}^{L}\frac{\partial h_i}{\partial h_{i-1}}\cdot\frac{\partial h_l}{\partial W_l} \tag{16}$$

where h_i represents the output of the ith layer. Skip connections introduce an additional term, effectively modifying the gradient propagation:

$$\frac{\partial\mathcal{L}}{\partial W_l}=\left(\prod_{i=l+1}^{L}\frac{\partial h_i}{\partial h_{i-1}}+I\right)\cdot\frac{\partial h_l}{\partial W_l} \tag{17}$$

where I represents the identity matrix that contributes to the gradient from the skipped layers. This matrix I ensures that at least part of the gradient from deeper layers (output end of the network) is transmitted without decay to earlier layers (input end of the network). This mechanism prevents the exponential decay of gradients and also enhances the training convergence. It allows earlier layers to benefit directly from both the raw error signal and the processed signals from subsequent layers, facilitating more effective learning throughout the network. □

Proposition 1. *The architectural optimizations within FUSERNet, particularly its reduced computational complexity and improved gradient stability, render it exceptionally suitable for deployment in environments where computational resources and memory are scarce.*

Proof. The combined effects of reduced computational complexity through separable convolutions and the maintenance of gradient integrity through skip connections enable FUSERNet to operate efficiently in resource-constrained environments. This efficiency allows for high performance without sacrificing accuracy and necessitating excessive computational overhead, thereby making FUSERNet ideal for resource-constrained devices. □

5. Experimental evaluation and results

In the comprehensive evaluation of our proposed models – Modified MobileNetV2, FUSERNet, and FusionNet – both quantitative metrics and qualitative analyses were utilized to ascertain their performance in aerial disaster image classification. This section delves deeper into these models internal characteristics through weight distributions and t-SNE visualizations. These internal characteristics of the model provide insightful perspectives on their learning behaviors and feature representation capabilities. While quantitative data highlights efficiency and accuracy, qualitative insights from weight distributions and t-SNE plots, coupled with confusion matrix analysis, reveal the nuances of model training, feature representation, and classification accuracy.

5.1. Model training

All networks in this study are developed and evaluated within the same training framework to ensure uniform conditions and enable a fair comparison during the inference phase. The Keras deep learning framework is employed, offering compatibility with all models available in TensorFlow, which serves as the backend. To maintain consistency, a uniform image size 224 × 224 pixels is utilized across all networks whenever feasible. Exceptions are made for the proposed FUSERNet model, which specifically requires a slightly higher image size i.e., 240 × 240 pixels (a typical size for training CNNs). The detailed federated learning training parameters are shown in Table 2. Before augmentation and inclusion in the batch, each image is resized to the appropriate dimensions depending on the network's requirements. It is worth noting that while larger image sizes are possible, they come at the expense of slower inference times. However, in this study, the exploration of different image sizes is not undertaken, with the primary focus placed on network design. Convergence details for the proposed models, along with a brief explanation of the underlying reasons for their performance characteristics, are provided in Table 3.

The initial phase of the training procedure involves the division of the dataset into training and test sets. The majority of the data is allocated to the training set, with the remainder test sets in a ratio of 80% and 20%, respectively. It is worth noting that the Normal class, being the majority class, is over-represented in the AIDER dataset, mirroring real-world conditions. However, this imbalance, if unaddressed, can potentially lead to overfitting issues where the network tends to classify everything as the majority class. To mitigate problems stemming from dataset imbalance, a strategy involving the simultaneous use of majority class undersampling and minority class oversampling within the same batch is implemented [18]. This ensures the selection of an equal number of images from each class to compose

Table 2
Federated learning training parameters.

Parameter	Value/Description
Number of clients	10
Clients selected per round	10% of total clients
Local epoch	10
Global rounds	300
Batch size	32
Learning rate	0.01
Optimizer	Stochastic Gradient Descent (SGD)
Loss Function	Sparse Categorical Crossentropy
Image size for models	224 × 224 pixels generally, 240 × 240 pixels for FUSERNet
Framework	Keras with TensorFlow backend

a batch, thereby ensuring a balanced representation of all cases. This approach is preferred not only to address class imbalance but also a significant portion of the images is extracted from videos. Even when videos are sparsely sampled, there exists temporal dependence between images in close proximity. Consequently, through random sampling, encounters with visually similar images during the training process are less frequent.

5.2. Computational infrastructure and software environment

In the training phase, our computational infrastructure is anchored on a standard desktop computer equipped with robust specifications, including an Intel Core i9-12900F CPU operating at 5 GHz, 64 GB RAM, a 500 GB SSD, and a 4 TB HDD. Additionally, an Nvidia RTX A5000 GPU with 24 GB of memory is employed to accelerate the training process. The software stack is built upon Python 3.9 running on Ubuntu 20.04, utilizing CUDA 11.4 and cuDNN 8.2 to harness GPU acceleration for enhanced performance. To simulate real-world deployment, we used Node.js for real-time client–server communication through socket programming, ensuring seamless interaction between UAVs and fog devices. The server program runs on a fog device, while the client operates on UAVs.

5.3. Quantitative performance analysis

We commenced our evaluation by applying Federated Transfer Learning to the pre-trained CNN architectures as detailed in Section 4.1. The summarized outcomes of this evaluation are presented in Table 4, providing a comparative analysis across various CNN-based models tailored for aerial disaster image classification. These models were assessed based on critical metrics such as parameter count, memory usage, and classification accuracy. Models like NASNetMobile and VGG16 demonstrated high accuracies; however, they require substantial memory resources which is not viable for UAVs due to their computational capacity. Conversely, EmergencyNet offers an exemplary case of memory efficiency, crucial for deployment in memory-restricted devices but lacks in sophisticated accuracy. Our proposed models, Modified MobileNetV2, FUSERNet, and FusionNet, highlighted remarkable efficiencies with minimal parameter counts and reduced memory footprints, achieving accuracies of 97.68%, 97.47%, and 97.75% respectively on AIDER Dataset. However, the reduction in accuracy on the NDD dataset compared to the AIDER dataset arise from the increased complexity of distinguishing features in natural disaster images. In contrast to the AIDER dataset, which features well-defined urban elements such as collapsed buildings and fires with localized, prominent patterns, the NDD dataset focuses on natural disasters with more spatially diffuse and less distinct patterns, like cyclones. The subtle variations in textures, shapes, and contextual clues across these expansive landscapes present significant challenges for feature extraction, particularly for CNN-based models, which rely heavily on localized features for effective classification. Additionally, the NDD dataset exhibits greater

Table 3
Proposed models convergence details.

Model name	Initial setup	No. of clients	Federated rounds for convergence	Reason
Modified MobileNetV2	Pre-trained	10% of total clients	<50	Quick convergence due to pre-trained weights
FUSERNet	From Scratch	10% of total clients	<100	Longer to converge due to initial learning phase
FusionNet	Model 1 Pre-trained + Model 2 From Scratch	10% of total clients	<50	Efficient convergence leveraging prior learning

Table 4
Performance comparison of existing models with our proposed models on NDD and AIDER datasets.

S.no	Model	Parameters (M)	Memory (MB)	Accuracy (%)	
				NDD	AIDER
1	NaturalNet [21]	34.74	132.54	97.15	98.03
2	InceptionV3 [28]	21.81	83.21	90.12	91.23
3	VGG16 [26]	14.72	56.14	94.32	96.49
4	DenseNet121 [27]	7.04	26.87	93.90	95.00
5	EfficientNetB0 [29]	4.38	16.70	85.38	84.36
6	NASNetMobile [31]	4.27	16.31	97.71	98.70
7	MobileNetV2 [33]	2.26	9.20	95.88	97.40
8	MobileNetV3 [34]	0.94	3.59	67.03	63.96
9	MFEMANet [20]	0.14	0.55	96.13	97.00
10	EmergencyNet [18]	0.09	0.36	91.20	92.98
11	FusionNet (Ours)	2.32	8.86	96.91	97.75
12	Modified MobileNetV2 (Ours)	2.26	8.64	96.29	97.68
13	FUSERNet (Ours)	0.06	0.23	96.15	97.47

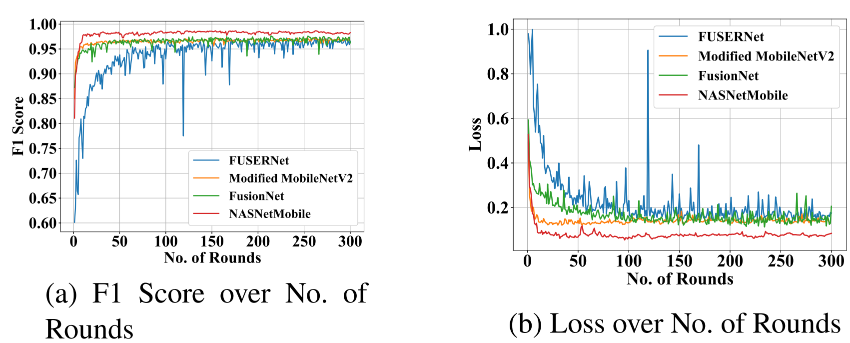


Fig. 9. F1 score and loss comparison with state-of-the-art model: (a) shows the F1 score over the number of rounds and (b) shows the loss over the number of rounds.

intra-class variability due to diverse geographic settings and varying lighting conditions, introducing noise that can impair the model's ability to generalize. Despite these challenges, our models are performing exceptionally well, competing with state-of-the-art models.

5.4. Evaluation of loss and F1 score

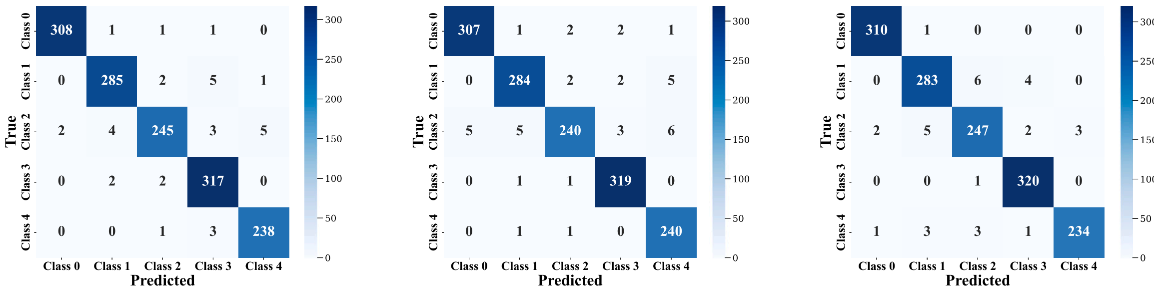
The loss and F1 scores offer additional performance metrics that are crucial for assessing the models' training efficiency and classification precision. Fig. 9(a) compares the F1 scores, with FusionNet achieving the highest scores, which reflects its superior capability to balance precision and recall effectively. Fig. 9(b) displays the loss comparison, showing that FusionNet consistently had the lowest loss values across all training rounds, indicating efficient learning and convergence.

5.5. Detailed evaluation of classification performance using confusion matrices

The confusion matrices detailed in Fig. 10 provide a succinct assessment of the classification performance of three models – Modified MobileNetV2, FUSERNet, and FusionNet – across diverse disaster scenarios including Fire, Flooded Areas, Normal, Traffic Accidents, and Collapsed Buildings. Modified MobileNetV2 is notably effective at classifying Fire and Flooded Areas, demonstrating minimal misclassifications in less relevant categories. FUSERNet shows a prominent accuracy in identifying Traffic Accidents and Normal scenarios, leveraging its self-attention mechanisms to better interpret urban environments. FusionNet combines the strengths of the aforementioned models to deliver comprehensive and balanced accuracy across all classes, showcasing the utility of model integration in enhancing disaster response capabilities. These analyses are crucial for understanding the practical implications of model performance in real-world applications, aiding in the ongoing refinement of disaster management technologies.

5.6. Performance analysis under varied environmental conditions

To assess the robustness of our models under varying environmental conditions, we conducted a series of tests to simulate low-light and high-traffic scenarios. These experiments were designed to identify potential performance degradation and latency increases that could affect the real-time capabilities of UAV-based applications. Table 5 presents the accuracy of each model before and after adjustments for low light and high traffic, highlighting the resilience and adaptability of our proposed models. Table 6 details the latency measurements, providing insights into the real-time processing potential of the models under different environmental stresses. These results were primarily obtained



(a) Modified MobileNetV2 (b) FUSERNet (c) FusionNet

Fig. 10. The confusion matrix for the proposed models illustrates the classification performance on the AIDER dataset. Class 0 represents Fire, Class 1 represents Traffic Accidents, Class 2 represents Normal, Class 3 represents Collapsed Buildings, and Class 4 represents Flooded Areas.

Table 5
Accuracy of models under varied environmental conditions.

Model	Low light		High traffic	
	Before	After	Before	After
NASNetMobile	98.70	98.70	98.70	97.54
MobileNetV2	97.40	97.26	97.40	94.31
Modified MobileNetV2	97.68	97.47	97.68	94.74
FusionNet (Ours)	97.75	97.53	97.75	96.35
FUSERNet (Ours)	97.47	97.21	97.47	96.49

Table 6
Latency of models under varied environmental conditions.

Model	Low light		High traffic	
	Before	After	Before	After
NASNetMobile	5.50	5.60	5.60	5.70
MobileNetV2	1.96	1.96	1.96	1.98
Modified MobileNetV2	1.96	1.97	1.96	1.97
FusionNet (Ours)	3.56	3.58	3.50	3.53
FUSERNet (Ours)	2.08	2.08	2.08	2.12

using the AIDER dataset, ensuring relevant and reliable performance metrics for UAV-based disaster response scenarios.

It is observed that the slight decrease in accuracy and the modest increase in latency for our proposed models – Modified MobileNetV2, FUSERNet, and FusionNet – highlight their substantial adaptability to complex environments. Increased latency under low light and high traffic conditions is attributed to the augmented computational demands required to maintain performance standards in these challenging settings. In low-light scenarios, our models intensify processing to compensate for reduced visibility, which inherently extends the time to accurately analyze data. Similarly, high-traffic conditions increase the complexity and density of the information to be processed, further amplifying the computational load. These adjustments are vital for ensuring that accuracy and reliability are not compromised under adverse conditions. Such calibrations are crucial for UAVs engaged in disaster response operations, where the ability to adapt to rapidly changing environments is essential. The modest increase in response times is a considered trade-off, prioritizing comprehensive analysis and reliable performance over mere processing speed, thereby it shows the robustness of our proposed models in practical deployment scenarios.

5.7. Qualitative insights from weight distributions

To further understand the internal dynamics and the generalization capabilities of the FUSERNet, weight distributions across various layers were examined. These distributions are crucial for evaluating the training stability and the potential of models to generalize across unseen data.

- **Initial Convolution Layer** (Fig. 11(a)): Shows a distribution with a peak around zero, suggesting a concentration of weights around lower values, which aids in detail preservation in image features without amplifying noise.
- **Dense Layer** (Fig. 11(b)): Demonstrated a Gaussian distribution, essential for the stability in the deeper network sections, crucial for robust feature synthesis before the final classification stage.
- **Separable Convolution Layer** (Fig. 11(c)): Exhibited a nearly normal distribution, indicative of effective regularization strategies preventing overfitting.

5.8. Feature space visualization with t-SNE

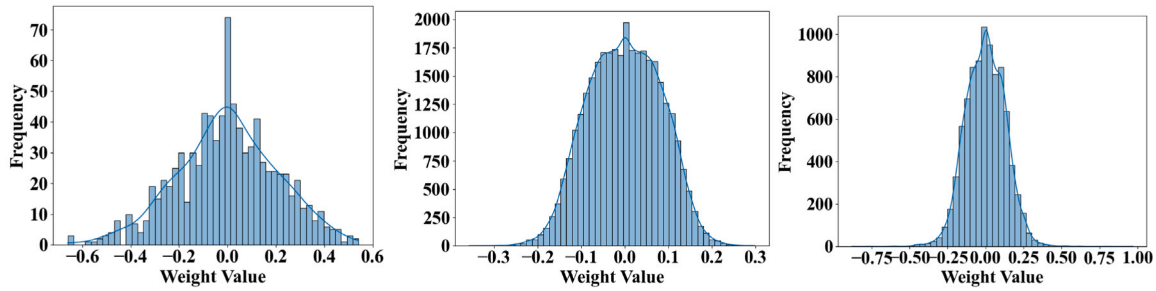
The t-SNE visualization, particularly for FUSERNet (Fig. 12), provides a vivid depiction of how distinct classes in the AIDER dataset such as Collapsed Buildings, Fire, Flooded Areas, Normal, and Traffic Accidents are spatially represented within the model's feature space. Effective clustering of similar instances alongside clear separation between different categories underscores the model's capability in discriminating and classifying complex image patterns.

5.9. Scalability analysis

To evaluate the scalability of our proposed models – Modified MobileNetV2, FUSERNet, and FusionNet – within a federated learning framework, we assessed their performance in relation to the increase in the number of participating clients. We experimented with client participation rates of 20%, 30%, and 40% of the total available clients, as detailed in Table 7. In the case of Modified MobileNetV2 and FusionNet, the analysis revealed a noticeable trend where increasing client participation to 30% enhanced model accuracy, likely due to the inclusion of a more diverse and representative sample of data in the training process. However, further increasing the participation to 40% led to a slight decline in accuracy, which could be attributed to the increased noise and variability in the larger set of data samples. For FuserNet, there was a decrease in accuracy as more clients were added, indicating potential issues with the dilution of essential information when averaging too many client updates. This analysis highlights the critical need to optimize the number of clients in federated learning to achieve a balance between data diversity and model reliability, essential for deploying robust models in diverse real-world scenarios.

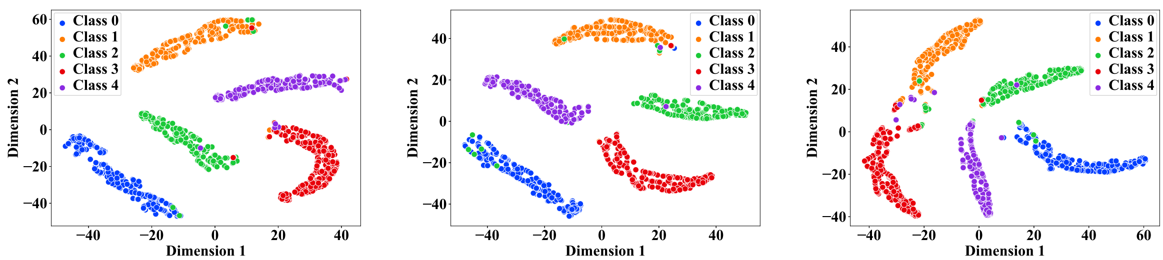
5.10. Comprehensive evaluation of proposed models efficacy

To thoroughly assess and compare the performance of our proposed models – MobileNetV2, FUSERNet, and FusionNet – across a variety of conditions and datasets, we have evaluated the sensitivity, specificity, false positive rate (FPR), F1-score, and accuracy. These metrics are



(a) Initial Conv2D layer (b) Dense Layer (c) Separable Convolution Layer

Fig. 11. The weight distributions in three different layers of the FUSERNet model are presented as follows: (a) shows the distribution of weights in the Initial Conv2D layer, (b) represents the weights in the Dense layer, and (c) depicts the weights in the Separable Convolution layer.



(a) Modified MobileNetV2 (b) FUSERNet (c) FusionNet

Fig. 12. t-SNE visualization of classification performance of the proposed models. The data points are colored according to their respective classes: Class 0 (Fire), Class 1 (Traffic Accidents), Class 2 (Normal), Class 3 (Collapsed Buildings), and Class 4 (Flooded Areas). (For interpretation of the references to color in this figure legend, the reader is referred to the web version of this article.)

Table 7
Scalability of model performance on change in number of clients.

Model name	NDD			AIDER		
	Number of clients			Number of clients		
	20%	30%	40%	20%	30%	40%
Modified MobileNetV2	96.29	96.35	96.10	97.68	97.82	97.37
FuserNet	95.15	94.91	95.10	97.47	97.34	97.23
FusionNet	96.91	96.93	96.82	97.75	97.81	97.51

essential for thoroughly understanding each model's capacity to accurately classify and distinguish between classes in diverse operational scenarios. Table 8 outlines the performance metrics for each model.

Sensitivity measures the model's ability to correctly identify positive instances, essential for scenarios where missing a true positive can have critical consequences. Specificity assesses how well the model can identify negative instances, helping to avoid false alarms that could divert resources in real-time operations. The FPR offers insights into the occurrence rate of false positives, while the F1-score balances the trade-offs between precision and recall, a critical measure in unbalanced datasets like those often encountered in disaster management. Finally, accuracy gives a straightforward indication of overall performance across all classes, ensuring the models perform well under various operational conditions. As illustrated in Table 8, FusionNet exhibits superior selectivity and sensitivity, highlighting its robustness in handling complex image classification tasks in dynamically changing environments. Moreover, the accuracy scores reaffirm the models' effectiveness, with FusionNet consistently demonstrating high accuracy, followed closely by FUSERNet and MobileNetV2.

5.11. Interpretability analysis using LIME

To further validate the effectiveness and interpretability of our federated learning models – MobileNetV2, FUSERNet, and FusionNet – used in UAV-based aerial image classification for disaster response, we employed Local Interpretable Model-agnostic Explanations (LIME). This method elucidates the decision-making process of our models at the instance level by highlighting the regions within the images that significantly influence the classification outcomes.

Fig. 13 illustrates the LIME results, where each model's focal points within a fire-impacted scenario are distinctly outlined. The image used for this analysis was taken from the Natural Disaster Dataset (NDD). These marked boundaries indicate regions of high relevance as determined by each model, shedding light on the discriminative features recognized by the models in classifying the scenes.

- **MobileNetV2:** This model primarily highlights the intense fire regions, capturing the bright flames which are key indicators of the fire class.
- **FUSERNet:** Similar to MobileNetV2, FUSERNet focuses on the fiery regions but extends its attention to surrounding smoke and areas with partial visibility, providing a broader context in its decision-making process.
- **FusionNet:** As an integration of the features recognized by both MobileNetV2 and FUSERNet, FusionNet shows an expanded focus that encompasses not only the core fire regions but also peripheral areas potentially indicative of spreading flames or emergent hotspots.

These visualizations not only substantiate the robustness of our models in recognizing and interpreting various aspects of aerial disaster imagery but also highlight the transparency of their reasoning processes. Such interpretability is crucial for ensuring trust and reliability in UAV-assisted emergency response operations, where understanding the basis of model predictions can significantly influence strategic decision-making.

5.12. Visual insights into model decisions through Grad-CAM

Grad-CAM (Gradient-weighted Class Activation Mapping) is a powerful visualization tool that highlights the regions in an input image

Table 8
Performance metrics comparison for three models across NDD and AIDER datasets, highlighting sensitivity, specificity, F1-score, accuracy, and FPR to assess model robustness.

Model	Class	NDD					AIDER				
		Sensitivity	Specificity	FPR	F1-score	Accuracy	Sensitivity	Specificity	FPR	F1-score	Accuracy
MobileNetV2	0	0.9600	0.9892	0.0108	0.9628	0.9821	0.9904	0.9982	0.0018	0.9919	0.9965
	1	0.9591	0.9924	0.0076	0.9627	0.9863	0.9727	0.9938	0.0062	0.9744	0.9895
	2	0.9664	0.9907	0.0093	0.9628	0.9863	0.9459	0.9949	0.0051	0.9608	0.9860
	3	0.9698	0.9894	0.0106	0.9574	0.9863	0.9875	0.9891	0.0109	0.9754	0.9888
	4	0.9614	0.9920	0.0080	0.9672	0.9849	0.9835	0.9949	0.0051	0.9794	0.9930
FUSERNet	0	0.9714	0.9882	0.0118	0.9673	0.9842	0.9871	0.9955	0.0045	0.9856	0.9937
	1	0.9591	0.9907	0.0093	0.9591	0.9849	0.9693	0.9929	0.0071	0.9709	0.9881
	2	0.9552	0.9899	0.0101	0.9552	0.9835	0.9266	0.9966	0.0034	0.9543	0.9839
	3	0.9655	0.9910	0.0090	0.9593	0.9870	0.9938	0.9937	0.0063	0.9861	0.9937
	4	0.9555	0.9920	0.0080	0.9641	0.9835	0.9917	0.9899	0.0101	0.9717	0.9902
FusionNet	0	0.9714	0.9919	0.0081	0.9728	0.9870	0.9968	0.9973	0.0027	0.9936	0.9972
	1	0.9703	0.9916	0.0084	0.9667	0.9876	0.9659	0.9921	0.0079	0.9675	0.9867
	2	0.9701	0.9907	0.0093	0.9647	0.9870	0.9537	0.9914	0.0086	0.9574	0.9846
	3	0.9655	0.9935	0.0065	0.9655	0.9890	0.9969	0.9937	0.0063	0.9877	0.9944
	4	0.9674	0.9937	0.0063	0.9731	0.9876	0.9669	0.9975	0.0025	0.9770	0.9923

(a) Original Image (b) Modified MobileNetV2 (c) FUSERNet (d) FusionNet

Fig. 13. LIME visualizations of fire classification: (a) Original image, (b) MobileNetV2, (c) FUSERNet, and (d) FusionNet, highlighting regions influencing the detection of fire-impacted areas.

that are important for predictions from convolutional neural networks. This technique is crucial for understanding the decision-making process of our models – MobileNetV2, FUSERNet, and FusionNet – especially in time-critical applications such as disaster management. The Grad-CAM heatmaps, as shown in Fig. 14, provide visual explanations by highlighting the areas within the image that most influence the outputs of the convolutional layers. The image used for generating these heatmaps are taken from the AIDER dataset. By applying Grad-CAM, we can see which features within the "Fire" scenario are most appropriate to each model's classification decisions.

- For **MobileNetV2**, the heatmap primarily highlights the intense fire zones, indicating the model's focus on the most visually striking elements of the fire.
- **FUSERNet** shows a broader activation, suggesting that this model considers both the fire and surrounding areas, possibly capturing contextual information that aids in its decision-making.
- **FusionNet**, which integrates features from multiple perspectives, exhibits a distinct pattern, focusing intensely on the core areas of the fire while also paying attention to the peripheries, indicating a nuanced understanding of the scene.

These heatmaps are indispensable for confirming that the models are indeed focusing on relevant parts of the images to make their predictions. Visualizing these focus areas helps verify the reliability and robustness of the models in real-world applications, where accurate and explainable predictions are crucial.

5.13. Environmental impact analysis

In an effort to develop sustainable and environmentally conscious DL models, we conducted an energy efficiency and carbon emissions analysis for our proposed models—Modified MobileNetV2, FUSERNet, and FusionNet. This analysis compares our models with other established models, highlighting the balance between model accuracy and environmental impact. For each model, we calculated the energy consumption over one hour of operation in kilowatt-hours (kWh), derived from the power usage values in watts. Subsequently, carbon emissions ($kgCO_2$) were estimated using the following formula:

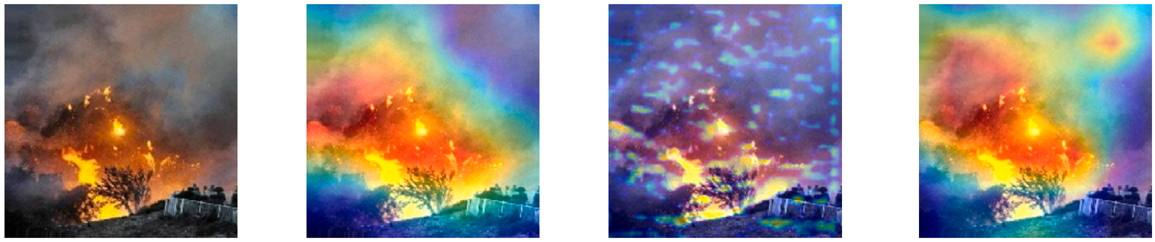
$$\text{Carbon Emissions } (kgCO_2) = \text{Energy (kWh)} \times 0.432 \tag{18}$$

where an emission factor of 0.432 $kgCO_2$ per kWh is applied, representing a standard estimate based on typical global energy sources [37].

As shown in Table 9, our models – Modified MobileNetV2, FUSERNet, and FusionNet – demonstrate a notable reduction in carbon emissions while maintaining high accuracy. FusionNet, for instance, achieves an accuracy of 97.75% with reduced carbon emissions, balancing high performance and sustainability. Similarly, Modified MobileNetV2 and FUSERNet achieve high accuracy levels of 97.68% and 97.47%, respectively, while exhibiting lower carbon footprints compared to traditional models such as DenseNet121 and VGG16. Although EmergencyNet achieves the lowest carbon emissions due to its minimal power requirements, its accuracy of 92.98% is lower than that of our proposed models, underscoring the trade-off between sustainability and performance. In contrast, our models achieve both competitive accuracy and reduced environmental impact, making them ideal models for sustainable, real-time UAV-based disaster response applications within the FU-Serve platform.

6. Conclusion

Our work shows a significant advancement in onboard UAV processing for emergency response applications, focusing on the unique challenges addressed by the FU-Serve platform. We conducted a thorough analysis, emphasizing the design and implementation of an efficient



(a) Original Image (b) Modified MobileNetV2 (c) FUSERNet (d) FusionNet

Fig. 14. Grad-CAM visualizations for different models analyzing a fire scene: (a) Original image, (b) MobileNetV2, (c) FUSERNet, and (d) FusionNet. These heatmaps highlight the critical regions influencing the models' predictions, thereby demonstrating each model's focus and interpretability in analyzing disaster scenarios.

Table 9
Model energy consumption and estimated carbon emissions.

Model name	Energy (kWh)	Carbon emissions (kgCO_2)
NASNetMobile	0.203	0.088
EfficientNetB0	0.201	0.087
VGG16	0.198	0.086
MobileNetV2	0.184	0.080
DenseNet121	0.179	0.077
InceptionV3	0.176	0.076
FusionNet (Ours)	0.166	0.072
MobileNetV3	0.157	0.068
Modified MobileNetV2	0.150	0.065
FUSERNet (Ours)	0.135	0.058
EmergencyNet	0.121	0.052

deep learning system capable of autonomously recognizing and classifying disaster events in real-time directly from a UAV. The proposed FUSERNet balances accuracy, and model complexity, establishing it as a foundational component for similar use cases with constraints. Through rigorous experimentation, our Federated Transfer Learning approach, leveraging MobileNetV2, surpasses existing models, showcasing its efficacy in resource-constrained environments with an impressive accuracy of 97.68%. Notably, our newly introduced FUSERNet demonstrates remarkable efficiency, providing a 34× model size improvement, a significantly reduced memory footprint, and comparable and good accuracy compared to state-of-the-art models. This efficiency is crucial for on-board processing in UAVs with limited computational resources. Furthermore, the FusionNet model, a harmonious blend of FUSERNet and MobileNetV2, attains an impressive nearly 97.75% accuracy with only a modest increase in model size. This achievement highlights the synergistic potential of combining tailored features and transfer-learning-based approaches, furthering the capabilities of on-board UAV processing. Our collective evidence underscores the robustness and effectiveness of our proposed methodologies, showcasing their potential for UAV applications, especially in emergency scenarios where real-time, on-board decision-making is primal. In the future, we aim to optimize the proposed models for greater computational efficiency while maintaining accuracy. We also plan to integrate UAV-based human detection for real-time disaster assessment. Leveraging UAV-based fog nodes for on-site processing will reduce latency and enhance decision-making. Additionally, expanding the dataset with diverse disaster scenarios and multimodal data will improve model robustness. Finally, we propose a collaborative resource management framework among UAVs to optimize task allocation and enhance FUServe's operational efficiency.

CRediT authorship contribution statement

Raju Imandi: Writing – review & editing, Writing – original draft, Validation, Software, Resources, Methodology, Formal analysis, Data curation, Conceptualization. **Arijit Roy:** Project administration, Methodology, Investigation. **Yong-Guk Kim:** Resources, Formal analysis, Conceptualization. **Pavan Kumar B.N.:** Writing – review & editing, Supervision, Project administration, Investigation, Formal analysis.

Declaration of competing interest

The authors declare the following financial interests/personal relationships which may be considered as potential competing interests: Pavan Kumar B N reports was provided by Indian Institute of Information Technology Sri City. Pavan Kumar B N reports a relationship with Indian Institute of Information Technology Sri City that includes: employment. Pavan Kumar B N has patent pending to NA. NIL If there are other authors, they declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Data availability

No data was used for the research described in the article.

References

[1] P. Petrides, C. Kyrkou, P. Kolios, T. Theocharides, C. Panayiotou, Towards a holistic performance evaluation framework for drone-based object detection, in: 2017 International Conference on Unmanned Aircraft Systems, ICUAS, IEEE, 2017.

[2] A. Al-Kaff, D. Martín, F. García, A.d.l. Escalera, J. María Armingol, Survey of computer vision algorithms and applications for unmanned aerial vehicles, Expert Syst. Appl. 92 (2018) 447–463.

[3] J. Yapp, R. Seker, R. Babiceanu, UAV as a service: Enabling on-demand access and on-the-fly re-tasking of multi-tenant UAVs using cloud services, in: The 35^{Th} IEEE/AIAA Digi. Avios. Sys. Conf., IEEE, 2016, pp. 1–8.

[4] N. Pathak, S. Misra, A. Mukherjee, A. Roy, A.Y. Zomaya, UAV virtualization for enabling heterogeneous and persistent uav-as-a-service, IEEE Tras. Veh. Tech. 69 (6) (2020) 6731–6738.

[5] A. Hazra, A. Kalita, M. Gurusamy, Meeting the requirements of internet of things: The promise of edge computing, IEEE Internet Things J. 11 (5) (2024) 7474–7498.

[6] I. Raju, A. Roy, FU-serve: Fog-enabled UAV-as-a-service for IoT applications, in: GLOBECOM 2023 - 2023 IEEE Global Communications Conference, IEEE, 2023.

[7] A. Hazra, M. Adhikari, D.K. Sah, T. Amgoth, Fair scheduling and computation co-offloading for industrial applications in fog networks, IEEE Trans. Netw. Serv. Manag. 21 (2) (2024) 1867–1876.

[8] A. Gupta, S.K. Gupta, Unmanned aerial vehicles in collaboration with fog computing network for improving quality of service, Int. J. Commun. Syst. 37 (9) (2024).

[9] B. Sedlak, I. Murturi, P.K. Donta, S. Dustdar, A privacy enforcing framework for data streams on the edge, IEEE Trans. Emerg. Top. Comput. (2024) 1–12.

[10] Y. Li, Q. Zhang, X. Wang, R. Zeng, H. Li, I. Murturi, S. Dustdar, M. Huang, Federated learning for internet of things, in: Learning Techniques for the Internet of Things, Springer Nature Switzerland, 2023, pp. 33–55.

[11] Y. Qu, C. Dong, J. Zheng, H. Dai, F. Wu, S. Guo, A. Anpalagan, Empowering edge intelligence by air-ground integrated federated learning, IEEE Netw. 35 (5) (2021) 34–41.

[12] M. Fu, Y. Shi, Y. Zhou, Federated learning via unmanned aerial vehicle, IEEE Trans. Wirel. Commun. 23 (4) (2024) 2884–2900.

[13] C.T. Nguyen, N. Van Huynh, N.H. Chu, Y.M. Saputra, D.T. Hoang, D.N. Nguyen, Q.-V. Pham, D. Niyato, E. Dutkiewicz, W.-J. Hwang, Transfer learning for wireless networks: A comprehensive survey, Proc. IEEE 110 (8) (2022) 1073–1115.

[14] A. Roy, P. Bouvry, Opti-U: Optimal UAV selection for enabling UAV-as-a-service, in: IEEE Int. Conf. on Comm., IEEE, 2022, pp. 1–7.

[15] V. Bekbulatova, A. Morichetta, S. Dustdar, FL-SERENADE: Federated learning for semi-supervised network anomaly detection. a case study, in: 2023 IEEE Intl Conf on Dependable, Autonomic and Secure Computing, Intl Conf on Pervasive Intelligence and Computing, Intl Conf on Cloud and Big Data Computing, Intl Conf on Cyber Science and Technology Congress, DASC/PiCom/CBDCom/CyberSciTech, IEEE, 2023.

[16] O. Friha, M.A. Ferrag, L. Shu, L. Maglaras, K.-K.R. Choo, M. Nafaa, FELIDS: Federated learning-based intrusion detection system for agricultural internet of things, J. Parallel Distrib. Comput. 165 (2022) 17–31.

[17] R.S. Antunes, C.A. da Costa, A. Küderle, I.A. Yari, B. Eskofier, Federated learning for healthcare: Systematic review and architecture proposal, ACM Trans. Intell. Syst. Technol. 13 (4) (2022) 1–23.

[18] C. Kyrkou, T. Theocharides, EmergencyNet: Efficient aerial image classification for drone-based emergency monitoring using atrous convolutional feature fusion, IEEE J. Sel. Top. Appl. Earth Obs. Remote. Sens. 13 (2020) 1687–1699.

[19] H.C. Reis, V. Turk, Detection of forest fire using deep convolutional neural networks with transfer learning approach, Appl. Soft Comput. 143 (2023) 110362.

[20] P. Bhadra, A. Balabantaray, A.K. Pasayat, MFEMANet: an effective disaster image classification approach for practical risk assessment, Mach. Vis. Appl. 34 (5) (2023).

[21] M.H. Bashir, M. Ahmad, D.R. Rizvi, A.A.A. El-Latif, Efficient CNN-based disaster events classification using UAV-aided images for emergency response application, Neural Comput. Appl. 36 (18) (2024) 10599–10612.

[22] M. Hadid, Z. Al-Qaysi, Q.M. Hussein, R.A. Aljanabi, I.R. Abdulqader, M.S. Suzani, W. Shir, Semantic image retrieval analysis based on deep learning and singular value decomposition, Appl. Data Sci. Anal. 2024 (2024) 17–31.

[23] E. Maggiori, Y. Tarabalka, G. Charpiat, P. Alliez, Convolutional neural networks for large-scale remote-sensing image classification, IEEE Trans. Geosci. Remote Sens. 55 (2) (2017) 645–657.

[24] B. McMahan, E. Moore, D. Ramage, S. Hampson, B.A. y Arcas, Communication-efficient learning of deep networks from decentralized data, in: Artificial Intelligence and Statistics, PMLR, 2017, pp. 1273–1282.

[25] R. Ribani, M. Marengoni, A survey of transfer learning for convolutional neural networks, in: 2019 32nd SIBGRAPI Conference on Graphics, Patterns and Images Tutorials, SIBGRAPI-T, IEEE, 2019.

[26] K. Simonyan, A. Zisserman, Very deep convolutional networks for large-scale image recognition, 2014, arXiv preprint arXiv:1409.1556.

[27] G. Huang, Z. Liu, L. Van Der Maaten, K.Q. Weinberger, Densely connected convolutional networks, in: Proceedings of the IEEE Conference on Computer Vision and Pattern Recognition, 2017, pp. 4700–4708.

[28] C. Szegedy, V. Vanhoucke, S. Ioffe, J. Shlens, Z. Wojna, Rethinking the inception architecture for computer vision, in: Proceedings of the IEEE Conference on Computer Vision and Pattern Recognition, 2016, pp. 2818–2826.

[29] M. Tan, Q. Le, Efficientnet: Rethinking model scaling for convolutional neural networks, in: International Conference on Machine Learning, PMLR, 2019, pp. 6105–6114.

[30] M. Tan, Q. Le, Efficientnetv2: Smaller models and faster training, in: International Conference on Machine Learning, PMLR, 2021, pp. 10096–10106.

[31] B. Zoph, V. Vasudevan, J. Shlens, Q.V. Le, Learning transferable architectures for scalable image recognition, in: Proceedings of the IEEE Conference on Computer Vision and Pattern Recognition, 2018, pp. 8697–8710.

[32] F.N. Iandola, S. Han, M.W. Moskewicz, K. Ashraf, W.J. Dally, K. Keutzer, SqueezeNet: AlexNet-level accuracy with 50x fewer parameters and < 0.5 MB model size, 2016, arXiv preprint arXiv:1602.07360.

[33] M. Sandler, A. Howard, M. Zhu, A. Zhmoginov, L.-C. Chen, Mobilenetv2: Inverted residuals and linear bottlenecks, in: Proceedings of the IEEE Conference on Computer Vision and Pattern Recognition, 2018, pp. 4510–4520.

[34] A. Howard, M. Sandler, G. Chu, L.-C. Chen, B. Chen, M. Tan, W. Wang, Y. Zhu, R. Pang, V. Vasudevan, et al., Searching for mobilenetv3, in: Proceedings of the IEEE/CVF International Conference on Computer Vision, 2019, pp. 1314–1324.

[35] M. Rahnemoonfar, T. Chowdhury, A. Sarkar, D. Varshney, M. Yari, R.R. Murphy, Floodnet: A high resolution aerial imagery dataset for post flood scene understanding, IEEE Access 9 (2021) 89644–89654.

[36] X. Zhu, J. Liang, A. Hauptmann, MSNet: A multilevel instance segmentation network for natural disaster damage assessment in aerial videos, in: Proceedings of the IEEE/CVF Winter Conference on Applications of Computer Vision, 2021, pp. 2023–2032.

[37] A. Lacoste, A. Luccioni, V. Schmidt, T. Dandres, Quantifying the carbon emissions of machine learning, 2019, arXiv preprint arXiv:1910.09700.